

UE1 Lab report

IN-BCN

BIS BB24

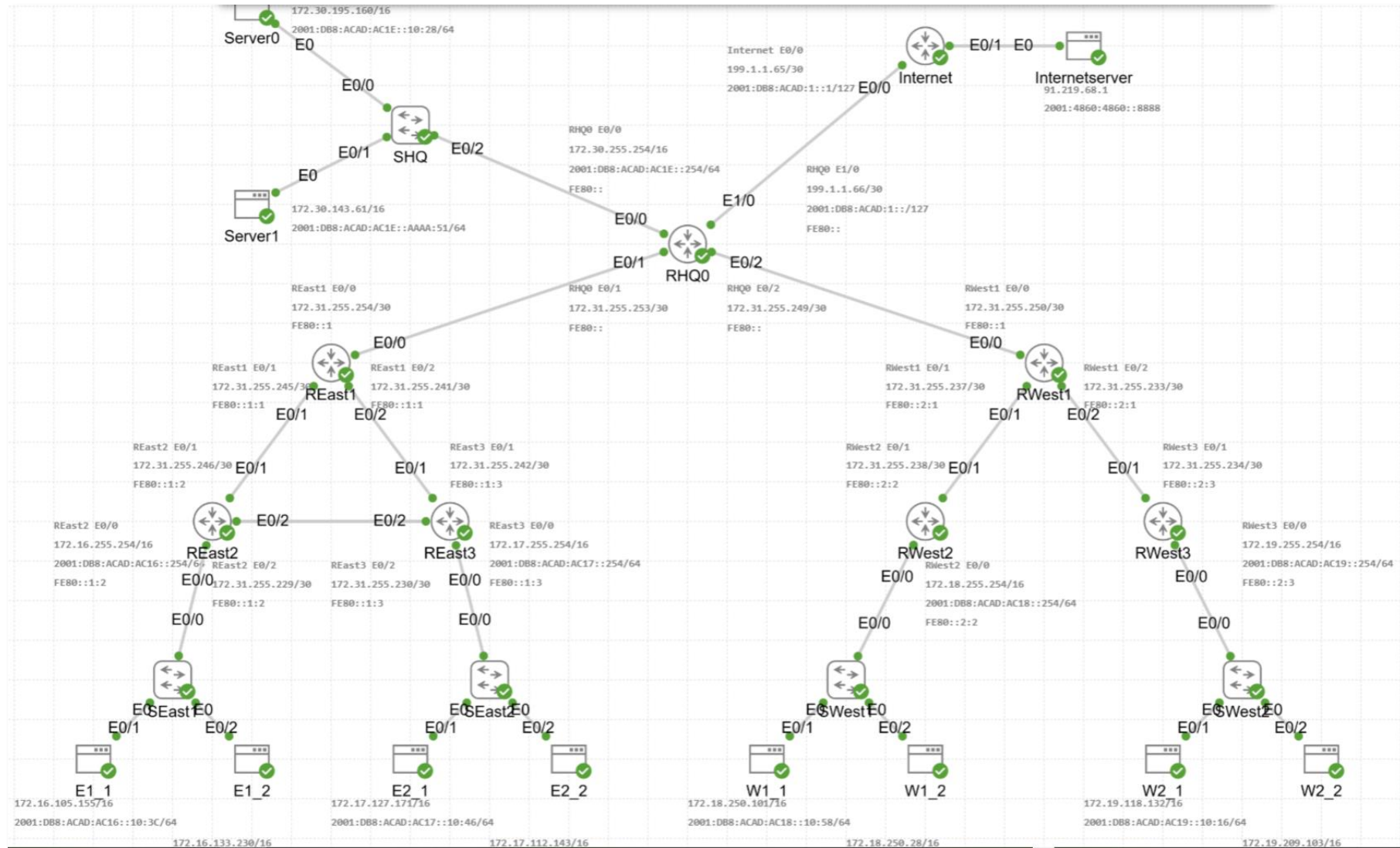
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Stand: 01. Juni 2025

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1 Exercise network plan



2 UE1.6) Analysis of RIP

a) Analyze and record the routing tables of routers RHQ0 and REast1

(Note: Codes are shown only the first time for better readability)

```
RHQ0# sh ip route
```

Codes: **L** - local, **C** - connected, **S** - static, **R** - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, **O** - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

```
S* 0.0.0.0/0 [1/0] via 199.1.1.65
R 172.16.0.0/16 [120/2] via 172.31.255.254, 00:00:17, Ethernet0/1
R 172.17.0.0/16 [120/2] via 172.31.255.254, 00:00:17, Ethernet0/1
R 172.18.0.0/16 [120/2] via 172.31.255.250, 00:00:04, Ethernet0/2
R 172.19.0.0/16 [120/2] via 172.31.255.250, 00:00:04, Ethernet0/2
172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.30.0.0/16 is directly connected, Ethernet0/0
L 172.30.255.254/32 is directly connected, Ethernet0/0
172.31.0.0/16 is variably subnetted, 9 subnets, 2 masks
R 172.31.255.228/30 [120/2] via 172.31.255.254, 00:00:17, Ethernet0/1
R 172.31.255.232/30 [120/1] via 172.31.255.250, 00:00:04, Ethernet0/2
R 172.31.255.236/30 [120/1] via 172.31.255.250, 00:00:04, Ethernet0/2
R 172.31.255.240/30 [120/1] via 172.31.255.254, 00:00:17, Ethernet0/1
R 172.31.255.244/30 [120/1] via 172.31.255.254, 00:00:17, Ethernet0/1
C 172.31.255.248/30 is directly connected, Ethernet0/2
L 172.31.255.249/32 is directly connected, Ethernet0/2
C 172.31.255.252/30 is directly connected, Ethernet0/1
L 172.31.255.253/32 is directly connected, Ethernet0/1
199.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 199.1.1.64/30 is directly connected, Ethernet1/0
L 199.1.1.66/32 is directly connected, Ethernet1/0
```

```
RHQ0# sh ipv6 route
```

```
S  ::/0 [1/0]
    via 2001:DB8:ACAD:1::1
C  2001:DB8:ACAD:1::/127 [0/0]
    via Ethernet1/0, directly connected
L  2001:DB8:ACAD:1::/128 [0/0]
    via Ethernet1/0, receive
R  2001:DB8:ACAD:AC16::/64 [120/3]
    via FE80::1, Ethernet0/1
R  2001:DB8:ACAD:AC17::/64 [120/3]
    via FE80::1, Ethernet0/1
R  2001:DB8:ACAD:AC18::/64 [120/3]
    via FE80::1, Ethernet0/2
R  2001:DB8:ACAD:AC19::/64 [120/3]
    via FE80::1, Ethernet0/2
C  2001:DB8:ACAD:AC1E::/64 [0/0]
    via Ethernet0/0, directly connected
L  2001:DB8:ACAD:AC1E::254/128 [0/0]
    via Ethernet0/0, receive
L  FF00::/8 [0/0]
    via Null0, receive
```

```
REast1# sho ip route
```

```
R*  0.0.0.0/0 [120/1] via 172.31.255.253, 00:00:15, Ethernet0/0
R   172.16.0.0/16 [120/1] via 172.31.255.246, 00:00:16, Ethernet0/1
R   172.17.0.0/16 [120/1] via 172.31.255.242, 00:00:04, Ethernet0/2
R   172.18.0.0/16 [120/3] via 172.31.255.253, 00:00:15, Ethernet0/0
R   172.19.0.0/16 [120/3] via 172.31.255.253, 00:00:15, Ethernet0/0
R   172.30.0.0/16 [120/1] via 172.31.255.253, 00:00:15, Ethernet0/0
    172.31.0.0/16 is variably subnetted, 10 subnets, 2 masks
R   172.31.255.228/30 [120/1] via 172.31.255.246, 00:00:16, Ethernet0/1
    [120/1] via 172.31.255.242, 00:00:04, Ethernet0/2
R   172.31.255.232/30 [120/2] via 172.31.255.253, 00:00:15, Ethernet0/0
R   172.31.255.236/30 [120/2] via 172.31.255.253, 00:00:15, Ethernet0/0
C   172.31.255.240/30 is directly connected, Ethernet0/2
L   172.31.255.241/32 is directly connected, Ethernet0/2
C   172.31.255.244/30 is directly connected, Ethernet0/1
L   172.31.255.245/32 is directly connected, Ethernet0/1
R   172.31.255.248/30 [120/1] via 172.31.255.253, 00:00:15, Ethernet0/0
C   172.31.255.252/30 is directly connected, Ethernet0/0
L   172.31.255.254/32 is directly connected, Ethernet0/0
```

```
REast1# sh ipv6 route
R  ::/0 [120/2]
    via FE80::, Ethernet0/0
R  2001:DB8:ACAD:AC16::/64 [120/2]
    via FE80::1:2, Ethernet0/1
R  2001:DB8:ACAD:AC17::/64 [120/2]
    via FE80::1:3, Ethernet0/2
R  2001:DB8:ACAD:AC18::/64 [120/4]
    via FE80::, Ethernet0/0
R  2001:DB8:ACAD:AC19::/64 [120/4]
    via FE80::, Ethernet0/0
R  2001:DB8:ACAD:AC1E::/64 [120/2]
    via FE80::, Ethernet0/0
L  FF00::/8 [0/0]
    via Null0, receive
```

→ From reviewing the tables, you can expect the RHQ0 router to be a hub router; it has Internet access and multiple point to point links.

REast1 is likely to be a branch router, because it relies on the default router of RHQ0.

Static routes (RHQ0) and RIP/ng are in use and can be seen in the tables, for the RIP/ng-learned routes, you can see the administrative distance and metric [ad=120/metric=1-4], in comparison, you can see [0/0] in ipv6 routing tables for the directly connected networks/interfaces: those will always be used first, because these are the lowest(=best) values for ad/metric.

a. How many routes were learned?

RHQ0 has learned 9 IPv4 RIP routes and 4 IPv6 routes,

REast1 has learned 10 IPv4 RIP routes (including the default route) and 6 IPv6 routes.

b) What is the administrative distance of RIP? Where can you find this value on the router?

Using the command

```
REast1# show ip router rip
```

...you can see the administrative distance for rip in the brackets next to the network, the second value is the metric within the same routing protocol.

```
R* 0.0.0.0/0 [120/1] via 172.31.255.253, 00:00:03, Ethernet0/0
```

Figure 1: Administrative distance of IPv4 default route, learned via RIP

c) How could you reduce the size of the routing tables?

Multiple options are available, for example:

- Using default routes
- Route aggregation / supernetting [1]
- Filtering route redistribution [2]
- Area based routing [3]

d) Which timers does RIP utilize?

“The routing information protocol uses the following timers as part of its operation: ...

- *Update Timer*
 - *The update timer controls the interval between two gratuitous Response Message. By default the value is 30 seconds. The response message is broadcast to all its RIP enabled interface.*
- *Invalid Timer*
 - *The invalid timer specifies how long a routing entry can be in the routing table without being updated. This is also called as expiration Timer. By default, the value is 180 seconds. After the timer expires the hop count of the routing entry will be set to 16, marking the destination as unreachable.*
- *Flush Timer*
 - *The flush timer controls the time between the route is invalidated or marked as unreachable and removal of entry from the routing table. By default the value is 240 seconds. This is 60 seconds longer than Invalid timer. So for 60 seconds the router will be advertising about this unreachable route to all its neighbours. This timer must be set to a higher value than the invalid timer.*
- *Hold-down Timer*

The hold-down timer is started per route entry, when the hop count is changing from lower value to higher value. This allows the route to get stabilized. During this time no update can be done to that routing entry. This is not part of the RFC 1058. This is Cisco's implementation. The default value of this timer is 180 seconds.”

(Timers, [4])

- e) How can the current values of these timers be displayed?

Using the command

```
show ip protocols
```

...you can see the configured timers and, in a table, the last update for learned networks.

```
REast1#sho ip prot
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 15 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
```

Figure 2: RIP timers

- f) Do some research on how to harden RIPng

Vulnerabilities of RIPng:

1. RIPng lacks built-in authentication mechanisms,
→ Packets can be spoofed or injected without verification
(RFC 2081, 3.4 [5])
2. Clear-Text Protocol: All routing information transmitted in plain text, relying on IPSec
(RFC 2080, 4 [6])
3. FF02::9 multicast is used for all updates (UDP 521)
→ Information leakage to unintended recipients
(RFC 2080, 2.1 [6])

Hardening mechanisms

1. IPSec Authentication
→ Configure IPv6 AH/ESP for packet authentication
(referred example vendor 'HUAWEI', product NE5000E router; [7])
2. Encryption using IPv6 integrated mechanisms (Next-Header-Field)
(referred example vendor 'HUAWEI', product NE40E-F universal service router; [8])
3. Configuring RIPng route control / using ACL
→ Filter RIPng traffic on UDP port 521 [9]
3. Distribute Lists
→ Control route advertisements with IPv6 prefix-lists [10]

- a. Ensure that RIPng updates are sent out only on point-to-point links only
This is configured using the following **command** on non-point-to-point interfaces:

```
router rip
passive-interface e0/0
```


3 UE1.8) Analysis OSPFv2

- a) Analyze the routing table of the routers RHQ0 and REast1. Which protocol is used to populate the entries in the routing table now?

RHQ0#sh ip route

```
S* 0.0.0.0/0 [1/0] via 199.1.1.65
R 172.16.0.0/16 [120/2] via 172.31.255.254, 00:00:09, Ethernet0/1
R 172.17.0.0/16 [120/2] via 172.31.255.254, 00:00:09, Ethernet0/1
R 172.18.0.0/16 [120/2] via 172.31.255.250, 00:00:00, Ethernet0/2
R 172.19.0.0/16 [120/2] via 172.31.255.250, 00:00:00, Ethernet0/2
172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.30.0.0/16 is directly connected, Ethernet0/0
L 172.30.255.254/32 is directly connected, Ethernet0/0
172.31.0.0/16 is variably subnetted, 9 subnets, 2 masks
R 172.31.255.228/30 [120/2] via 172.31.255.254, 00:00:09, Ethernet0/1
R 172.31.255.232/30 [120/1] via 172.31.255.250, 00:00:00, Ethernet0/2
R 172.31.255.236/30 [120/1] via 172.31.255.250, 00:00:00, Ethernet0/2
R 172.31.255.240/30 [120/1] via 172.31.255.254, 00:00:09, Ethernet0/1
R 172.31.255.244/30 [120/1] via 172.31.255.254, 00:00:09, Ethernet0/1
C 172.31.255.248/30 is directly connected, Ethernet0/2
L 172.31.255.249/32 is directly connected, Ethernet0/2
C 172.31.255.252/30 is directly connected, Ethernet0/1
L 172.31.255.253/32 is directly connected, Ethernet0/1
199.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 199.1.1.64/30 is directly connected, Ethernet1/0
L 199.1.1.66/32 is directly connected, Ethernet1/0
```

REast1#sh ip route

```
O*E2 0.0.0.0/0 [110/1] via 172.31.255.253, 00:37:13, Ethernet0/0
R 172.16.0.0/16 [120/1] via 172.31.255.246, 00:00:05, Ethernet0/1
R 172.17.0.0/16 [120/1] via 172.31.255.242, 00:00:22, Ethernet0/2
R 172.18.0.0/16 [120/3] via 172.31.255.253, 00:00:24, Ethernet0/0
R 172.19.0.0/16 [120/3] via 172.31.255.253, 00:00:24, Ethernet0/0
R 172.30.0.0/16 [120/1] via 172.31.255.253, 00:00:24, Ethernet0/0
172.31.0.0/16 is variably subnetted, 10 subnets, 2 masks
R 172.31.255.228/30 [120/1] via 172.31.255.246, 00:00:05, Ethernet0/1
[120/1] via 172.31.255.242, 00:00:22, Ethernet0/2
R 172.31.255.232/30 [120/2] via 172.31.255.253, 00:00:24, Ethernet0/0
R 172.31.255.236/30 [120/2] via 172.31.255.253, 00:00:24, Ethernet0/0
C 172.31.255.240/30 is directly connected, Ethernet0/2
L 172.31.255.241/32 is directly connected, Ethernet0/2
C 172.31.255.244/30 is directly connected, Ethernet0/1
L 172.31.255.245/32 is directly connected, Ethernet0/1
O 172.31.255.248/30 [110/20] via 172.31.255.253, 00:37:14, Ethernet0/0
C 172.31.255.252/30 is directly connected, Ethernet0/0
L 172.31.255.254/32 is directly connected, Ethernet0/0
```

→ For router REast1, default route and the point-to-point network between RHQ0 and RWest1 are new learned via **OSPF**.

In terms of the additional codes to the OSPF-“O”, there is helpful official cisco routing table documentation: [11]

- b) Analyze the OSPF topology table and the neighbor table of RHQ0 and REast1. (Hint: you will find the commands for that in the OSPF lecture slides)

```
RHQ0#sh ip ospf da
      OSPF Router with ID (1.1.0.0) (Process ID 10)

      Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#         Checksum Link count
1.1.0.0      1.1.0.0        8        0x80000004  0x00F4F1  2
1.1.1.1      1.1.1.1        18       0x8000000D  0x00AAD3  1
1.1.2.1      1.1.2.1        18       0x8000000A  0x003851  1

      Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#         Checksum
172.31.255.250 1.1.2.1      17       0x80000005  0x009ACB
172.31.255.254 1.1.1.1      17       0x80000005  0x0070F3

      Type-5 AS External Link States

Link ID      ADV Router    Age      Seq#         Checksum Tag
0.0.0.0      1.1.0.0      16       0x80000001  0x00CCDA  10
172.31.255.240 1.1.1.1      925      0x80000002  0x0012CF  0
172.31.255.244 1.1.1.1      925      0x80000002  0x00E9F3  0

RHQ0#sho ip ospf nei
Neighbor ID  Pri  State           Dead Time   Address      Interface
1.1.1.1      1    FULL/DR         00:00:37   172.31.255.254 Ethernet0/1
1.1.2.1      1    FULL/DR         00:00:33   172.31.255.250 Ethernet0/2
```

```
REast1#sho ip osp da
      OSPF Router with ID (1.1.1.1) (Process ID 10)

      Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum Link count
1.1.0.0      1.1.0.0      74       0x80000004   0x00F4F1 2
1.1.1.1      1.1.1.1      82       0x8000000D   0x00AAD3 1
1.1.2.1      1.1.2.1      83       0x8000000A   0x003851 1

      Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
172.31.255.250 1.1.2.1      83       0x80000005   0x009ACB
172.31.255.254 1.1.1.1      82       0x80000005   0x0070F3

      Type-5 AS External Link States

Link ID      ADV Router    Age      Seq#          Checksum Tag
0.0.0.0      1.1.0.0      82       0x80000001   0x00CCDA 10
172.31.255.240 1.1.1.1      988      0x80000002   0x0012CF 0
172.31.255.244 1.1.1.1      988      0x80000002   0x00E9F3 0

REast1#sho ip os nei
Neighbor ID  Pri  State           Dead Time   Address      Interface
1.1.0.0      1    FULL/BDR        00:00:34   172.31.255.253 Ethernet0/0
```

→ Both routers have the same Router Link States(Router IDs: 1.1.0.0 for RHQ0, 1.1.1.1 for REast1 and 1.1.2.1 for RWest1). **ADV Router** stands for advertising router (the one, that originated the LSA), **Seq#** indicates the most recent version of the LSA.

RHQ0 has 2 links (**Link count**) and REast1/RWest1 both have 1, which indicates, that they are connected to RHQ0. This is also confirmed with the show neighbor commands.

In the neighbor table of RHQ0, you can see that both branch routers were elected **D**esignated **R**outers (and RHQ0 **B**ackup **D**R), this is because of the higher router ID; priority (**Pri**) is 1 (default) on all routers, so the router ID is, what counts in the election process (Page 17 [12]).

4 UE1.10) Analysis OSPFv3

a) Analyze the routing table of the routers RHQ0 and REast1. Which protocol is used to populate the entries in the routing table now?

(Note: added IPv4 routing table for completeness)

```
RHQ0#sh ip rou
S* 0.0.0.0/0 [1/0] via 199.1.1.65
    172.16.0.0/16 [120/2] via 172.31.255.254, 00:00:11, Ethernet0/1
    172.17.0.0/16 [120/2] via 172.31.255.254, 00:00:11, Ethernet0/1
    172.18.0.0/16 [120/2] via 172.31.255.250, 00:00:11, Ethernet0/2
    172.19.0.0/16 [120/2] via 172.31.255.250, 00:00:11, Ethernet0/2
    172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
        172.30.0.0/16 is directly connected, Ethernet0/0
        172.30.255.254/32 is directly connected, Ethernet0/0
    172.31.0.0/16 is variably subnetted, 9 subnets, 2 masks
        172.31.255.228/30 [120/2] via 172.31.255.254, 00:00:11, Ethernet0/1
        172.31.255.232/30 [120/1] via 172.31.255.250, 00:00:11, Ethernet0/2
        172.31.255.236/30 [120/1] via 172.31.255.250, 00:00:11, Ethernet0/2
        172.31.255.240/30 [120/1] via 172.31.255.254, 00:00:11, Ethernet0/1
        172.31.255.244/30 [120/1] via 172.31.255.254, 00:00:11, Ethernet0/1
        172.31.255.248/30 is directly connected, Ethernet0/2
        172.31.255.249/32 is directly connected, Ethernet0/2
        172.31.255.252/30 is directly connected, Ethernet0/1
        172.31.255.253/32 is directly connected, Ethernet0/1
    199.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
        199.1.1.64/30 is directly connected, Ethernet1/0
        199.1.1.66/32 is directly connected, Ethernet1/0
```

```
RHQ0#sh ipv rou
S  ::/0 [1/0]
    via 2001:DB8:ACAD:1::1
C  2001:DB8:ACAD:1::/127 [0/0]
    via Ethernet1/0, directly connected
L  2001:DB8:ACAD:1::/128 [0/0]
    via Ethernet1/0, receive
R  2001:DB8:ACAD:AC16::/64 [120/3]
    via FE80::1, Ethernet0/1
R  2001:DB8:ACAD:AC17::/64 [120/3]
    via FE80::1, Ethernet0/1
R  2001:DB8:ACAD:AC18::/64 [120/3]
    via FE80::1, Ethernet0/2
R  2001:DB8:ACAD:AC19::/64 [120/3]
    via FE80::1, Ethernet0/2
C  2001:DB8:ACAD:AC1E::/64 [0/0]
    via Ethernet0/0, directly connected
L  2001:DB8:ACAD:AC1E::254/128 [0/0]
    via Ethernet0/0, receive
L  FF00::/8 [0/0]
    via Null0, receive
```

```
REast1#sh ip rou
OE2 0.0.0.0/0 [110/1] via 172.31.255.253, 00:00:11, Ethernet0/0
    172.16.0.0/16 [120/1] via 172.31.255.246, 00:00:20, Ethernet0/1
    172.17.0.0/16 [120/1] via 172.31.255.242, 00:00:15, Ethernet0/2
    172.18.0.0/16 [120/3] via 172.31.255.253, 00:00:17, Ethernet0/0
    172.19.0.0/16 [120/3] via 172.31.255.253, 00:00:17, Ethernet0/0
    172.30.0.0/16 [120/1] via 172.31.255.253, 00:00:17, Ethernet0/0
    172.31.0.0/16 is variably subnetted, 10 subnets, 2 masks
    172.31.255.228/30 [120/1] via 172.31.255.246, 00:00:20, Ethernet0/1
    172.31.255.232/30 [120/1] via 172.31.255.242, 00:00:15, Ethernet0/2
    172.31.255.236/30 [120/2] via 172.31.255.253, 00:00:17, Ethernet0/0
    172.31.255.240/30 [120/2] via 172.31.255.253, 00:00:17, Ethernet0/0
    172.31.255.240/30 is directly connected, Ethernet0/2
    172.31.255.241/32 is directly connected, Ethernet0/2
    172.31.255.244/30 is directly connected, Ethernet0/1
    172.31.255.245/32 is directly connected, Ethernet0/1
    172.31.255.248/30 [110/20] via 172.31.255.253, 00:00:11, Ethernet0/0
    172.31.255.252/30 is directly connected, Ethernet0/0
    172.31.255.254/32 is directly connected, Ethernet0/0
```

```
REast1# sho ipv rou
OE2 ::/0 [110/1], tag 10
    via FE80::, Ethernet0/0
R 2001:DB8:ACAD:AC16::/64 [120/2]
    via FE80::1:2, Ethernet0/1
R 2001:DB8:ACAD:AC17::/64 [120/2]
    via FE80::1:3, Ethernet0/2
R 2001:DB8:ACAD:AC18::/64 [120/4]
    via FE80::, Ethernet0/0
R 2001:DB8:ACAD:AC19::/64 [120/4]
    via FE80::, Ethernet0/0
R 2001:DB8:ACAD:AC1E::/64 [120/2]
    via FE80::, Ethernet0/0
L FF00::/8 [0/0]
    via Null0, receive
```

→ On RHQ0, there is no difference (see above: 2 - UE1.6) Analysis of RIP – RHQ0 IPv6 routing table), because both branch routers do not redistribute routes yet. On REast1, the default route is now learned via OSPF, for additional routing table codes see [11].

5 UE1.12) Analysis Multiarea OSPF Dual Stack and Route Redistribution

a) Analyze the changes in the routing tables of RHQ0, RWest2 and RWest3.

RHQ0#sho ip rou

```
S* 0.0.0.0/0 [1/0] via 199.1.1.65
O E1 172.16.0.0/16 [110/110] via 172.31.255.254, 00:05:08, Ethernet0/1
O E1 172.17.0.0/16 [110/110] via 172.31.255.254, 00:05:08, Ethernet0/1
O E2 172.18.0.0/16 [110/20] via 172.31.255.250, 00:03:38, Ethernet0/2
O E2 172.19.0.0/16 [110/20] via 172.31.255.250, 00:03:28, Ethernet0/2
    172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.30.0.0/16 is directly connected, Ethernet0/0
L    172.30.255.254/32 is directly connected, Ethernet0/0
    172.31.0.0/16 is variably subnetted, 9 subnets, 2 masks
O E1 172.31.255.228/30 [110/110] via 172.31.255.254, 00:05:08, Ethernet0/1
O IA 172.31.255.232/30 [110/20] via 172.31.255.250, 00:04:30, Ethernet0/2
O IA 172.31.255.236/30 [110/20] via 172.31.255.250, 00:04:30, Ethernet0/2
O E2 172.31.255.240/30 [110/20] via 172.31.255.254, 00:05:22, Ethernet0/1
O E2 172.31.255.244/30 [110/20] via 172.31.255.254, 00:05:22, Ethernet0/1
C    172.31.255.248/30 is directly connected, Ethernet0/2
L    172.31.255.249/32 is directly connected, Ethernet0/2
C    172.31.255.252/30 is directly connected, Ethernet0/1
L    172.31.255.253/32 is directly connected, Ethernet0/1
    199.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    199.1.1.64/30 is directly connected, Ethernet1/0
L    199.1.1.66/32 is directly connected, Ethernet1/0
```

RHQ0#sho ipv rou

```
S ::/0 [1/0]
    via 2001:DB8:ACAD:1::1
C 2001:DB8:ACAD:1::/127 [0/0]
    via Ethernet1/0, directly connected
L 2001:DB8:ACAD:1::/128 [0/0]
    via Ethernet1/0, receive
OE1 2001:DB8:ACAD:AC16::/64 [110/110]
    via FE80::1, Ethernet0/1
OE1 2001:DB8:ACAD:AC17::/64 [110/110]
    via FE80::1, Ethernet0/1
OE2 2001:DB8:ACAD:AC18::/64 [110/20]
    via FE80::1, Ethernet0/2
OE2 2001:DB8:ACAD:AC19::/64 [110/20]
    via FE80::1, Ethernet0/2
C 2001:DB8:ACAD:AC1E::/64 [0/0]
    via Ethernet0/0, directly connected
L 2001:DB8:ACAD:AC1E::254/128 [0/0]
    via Ethernet0/0, receive
L FF00::/8 [0/0]
    via Null0, receive
```

```
RWest2#sho ip rou
```

```
0*E2 0.0.0.0/0 [110/1] via 172.31.255.237, 00:05:17, Ethernet0/1
0 E1 172.16.0.0/16 [110/130] via 172.31.255.237, 00:05:17, Ethernet0/1
0 E1 172.17.0.0/16 [110/130] via 172.31.255.237, 00:05:17, Ethernet0/1
    172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.18.0.0/16 is directly connected, Ethernet0/0
L    172.18.255.254/32 is directly connected, Ethernet0/0
0 E2 172.19.0.0/16 [110/20] via 172.31.255.237, 00:05:07, Ethernet0/1
    172.31.0.0/16 is variably subnetted, 8 subnets, 2 masks
0 E1 172.31.255.228/30 [110/130] via 172.31.255.237, 00:05:17, Ethernet0/1
0 IA 172.31.255.232/30 [110/20] via 172.31.255.237, 00:05:17, Ethernet0/1
C    172.31.255.236/30 is directly connected, Ethernet0/1
L    172.31.255.238/32 is directly connected, Ethernet0/1
0 E2 172.31.255.240/30 [110/20] via 172.31.255.237, 00:05:17, Ethernet0/1
0 E2 172.31.255.244/30 [110/20] via 172.31.255.237, 00:05:17, Ethernet0/1
0 IA 172.31.255.248/30 [110/20] via 172.31.255.237, 00:05:17, Ethernet0/1
0 IA 172.31.255.252/30 [110/30] via 172.31.255.237, 00:05:17, Ethernet0/1
```

```
RWest2#sho ipv rou
```

```
OE2 ::/0 [110/1], tag 10
    via FE80::2:1, Ethernet0/1
OE1 2001:DB8:ACAD:AC16::/64 [110/130]
    via FE80::2:1, Ethernet0/1
OE1 2001:DB8:ACAD:AC17::/64 [110/130]
    via FE80::2:1, Ethernet0/1
C    2001:DB8:ACAD:AC18::/64 [0/0]
    via Ethernet0/0, directly connected
OE2 2001:DB8:ACAD:AC19::/64 [110/20]
    via FE80::2:1, Ethernet0/1
L    2001:DB8:ACAD:AC18::254/128 [0/0]
    via Ethernet0/0, receive
L    FF00::/8 [0/0]
    via Null0, receive
```

RWest3#sho ip rou

```

O*E2 0.0.0.0/0 [110/1] via 172.31.255.233, 00:06:32, Ethernet0/1
O E1 172.16.0.0/16 [110/130] via 172.31.255.233, 00:06:32, Ethernet0/1
O E1 172.17.0.0/16 [110/130] via 172.31.255.233, 00:06:32, Ethernet0/1
O E2 172.18.0.0/16 [110/20] via 172.31.255.233, 00:06:32, Ethernet0/1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/0
L    172.19.255.254/32 is directly connected, Ethernet0/0
    172.31.0.0/16 is variably subnetted, 8 subnets, 2 masks
O E1 172.31.255.228/30 [110/130] via 172.31.255.233, 00:06:32, Ethernet0/1
C    172.31.255.232/30 is directly connected, Ethernet0/1
L    172.31.255.234/32 is directly connected, Ethernet0/1
O IA 172.31.255.236/30 [110/20] via 172.31.255.233, 00:06:32, Ethernet0/1
O E2 172.31.255.240/30 [110/20] via 172.31.255.233, 00:06:32, Ethernet0/1
O E2 172.31.255.244/30 [110/20] via 172.31.255.233, 00:06:32, Ethernet0/1
O IA 172.31.255.248/30 [110/20] via 172.31.255.233, 00:06:32, Ethernet0/1
O IA 172.31.255.252/30 [110/30] via 172.31.255.233, 00:06:32, Ethernet0/1

```

RWest3#sho ipv rou

```

OE2 ::/0 [110/1], tag 10
    via FE80::2:1, Ethernet0/1
OE1 2001:DB8:ACAD:AC16::/64 [110/130]
    via FE80::2:1, Ethernet0/1
OE1 2001:DB8:ACAD:AC17::/64 [110/130]
    via FE80::2:1, Ethernet0/1
OE2 2001:DB8:ACAD:AC18::/64 [110/20]
    via FE80::2:1, Ethernet0/1
C 2001:DB8:ACAD:AC19::/64 [0/0]
    via Ethernet0/0, directly connected
L 2001:DB8:ACAD:AC19::254/128 [0/0]
    via Ethernet0/0, receive
L FF00::/8 [0/0]
    via Null0, receive

```

→

Router	Default Route code	S IPv4/6	I IPv4/6	C IPv4/6	OE1 IPv4/6	OE2 IPv4/6	O(A) IPv4/6
RHQ0	S	1/1	4/3	4/2	3/2	4/2	2/0
RWest2	O*E2	0/0	2/2	2/1	3/2	4/2	3/0
RWest3	O*E2	0/0	2/2	2/1	3/2	4/2	3/0

The OE1 routes on all three routers are coming from the redistribution from East side with metric 100 and metric type 1, metric adds up (also, the internal OSPF cost to ASBR is added with type 1 – but not with type 2) and therefor West side routers have a metric of 130 to East side networks.

The interarea routes on RHQ0 point in the direction of RWest1, where the OSPF areas 1+2 are configured. For the branch networks behind RWest1+2, there are OE2 routes.

All routes should be able to reach the four client networks

172.16/17/18/19.x.x and 2001:DB8:ACAD:AC16/17/18/19.

6 UE1.14) Analysis Multiarea OSPF Dual Stack – Stub & Totally Stub Area

a) Analyze the routing tables of RWest2 and RWest3. What's the difference? Which entries have changed and why?

```
RWest2#sho ip rou
O*IA 0.0.0.0/0 [110/11] via 172.31.255.237, 00:00:27, Ethernet0/1
      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.18.0.0/16 is directly connected, Ethernet0/0
L      172.18.255.254/32 is directly connected, Ethernet0/0
      172.31.0.0/16 is variably subnetted, 5 subnets, 2 masks
O IA  172.31.255.232/30 [110/20] via 172.31.255.237, 00:00:27, Ethernet0/1
C      172.31.255.236/30 is directly connected, Ethernet0/1
L      172.31.255.238/32 is directly connected, Ethernet0/1
O IA  172.31.255.248/30 [110/20] via 172.31.255.237, 00:00:27, Ethernet0/1
O IA  172.31.255.252/30 [110/30] via 172.31.255.237, 00:00:27, Ethernet0/1

RWest2#sho ipv rou
OI ::/0 [110/11]
      via FE80::2:1, Ethernet0/1
C  2001:DB8:ACAD:AC18::/64 [0/0]
      via Ethernet0/0, directly connected
L  2001:DB8:ACAD:AC18::254/128 [0/0]
      via Ethernet0/0, receive
L  FF00::/8 [0/0]
      via Null0, receive
```

```
RWest3#sho ip rou
O*IA 0.0.0.0/0 [110/11] via 172.31.255.233, 00:01:31, Ethernet0/1
      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.19.0.0/16 is directly connected, Ethernet0/0
L      172.19.255.254/32 is directly connected, Ethernet0/0
      172.31.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.31.255.232/30 is directly connected, Ethernet0/1
L      172.31.255.234/32 is directly connected, Ethernet0/1

RWest3#sho ipv rou
OI ::/0 [110/11]
      via FE80::2:1, Ethernet0/1
C  2001:DB8:ACAD:AC19::/64 [0/0]
      via Ethernet0/0, directly connected
L  2001:DB8:ACAD:AC19::254/128 [0/0]
      via Ethernet0/0, receive
L  FF00::/8 [0/0]
      via Null0, receive
```

→

Previously (from UE1.12 analysis):

Router	Default Route code	 IPv4/6	 IPv4/6	 IPv4/6	 OE1 IPv4/6	 OE2 IPv4/6	 O(A) IPv4/6
RWest2	O*E2	0/0	2/2	2/1	3/2	4/2	3/0
RWest3	O*E2	0/0	2/2	2/1	3/2	4/2	3/0

Now (after area 1 = stub, area 2 = stub no-summary):

Router	Default Route code	 IPv4/6	 IPv4/6	 IPv4/6	 OE1 IPv4/6	 OE2 IPv4/6	 O(A) IPv4/6
RWest2	 O*IA	0/0	2/2	2/1	 0/0	 0/0	 4/1
RWest3	 O*IA	0/0	2/2	2/1	 0/0	 0/0	 1/1

RWest2 learns only LSA1-3 routes, hence, the three routes with metric 20 are still in the table.

In contrast, RWest3 does not learn routes from other areas, but only from its own area.

The ABR injects a default route in both cases (previously, this was an external route O*E2).

(pages 7-11 [13]; page 36 [14])

7 References

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